
COURSE CURRICULUM

**AI Awareness & Hands-on Workshop
for School Students**

Understanding AI, its impact, and its right use for learning and creating

Programme	AI Awareness & Hands-on Workshop for School Students
Offered by	Baderia Global Institute of Engineering & Management, Jabalpur
Target group	Students of Class 8 to Class 12
Duration	2 hours (one session)
Mode	Classroom session with live demonstration of AI tools
Resource person	Mr. Deepak Vishwakarma, Trainer, BGIEM

1. ABOUT THE PROGRAMME

This programme is conducted by Baderia Global Institute of Engineering & Management, Jabalpur, as part of its school outreach activities. It introduces school students to Artificial Intelligence in simple language: what it is, how it is changing our daily life and future careers, and how it can be used in the right way for studies and creative work.

Along with the basic concepts, students are shown some of the most useful AI tools available today, namely Gemini, Gemma, NotebookLM, Nano Banana, Google Flow, Lovable and Google Workspace Studio, through live demonstrations. Safe, honest and responsible use of AI is a part of every module.

2. COURSE OBJECTIVES

1. To give students a basic understanding of Artificial Intelligence and how it works.
2. To make students aware of the impact of AI on society, education and careers.
3. To introduce AI tools that help in learning, creating content, building apps and automating tasks.
4. To develop the habit of using AI safely, honestly and responsibly.

3. COURSE CONTENT

The course is divided into nine modules, and each module is further divided into chapters. The topics to be covered and the practical work or demonstration for each chapter are listed below. Practicals marked (*Demonstration*) are performed by the trainer on the projector, as the tool requires a paid plan or an organisational account.

Module	Chapter	Topics Covered	Practicals
MODULE 1 Introduction to Artificial Intelligence	1.1 What is Artificial Intelligence?	- Meaning and definition of Artificial Intelligence - Human intelligence and machine intelligence - A brief history of AI, from early computers to modern chatbots - AI in everyday life	- Identifying AI features in commonly used apps such as YouTube, Google Maps, Gmail and Google Lens “Real or AI?” activity on spotting AI-generated images
	1.2 How Machines Learn	- Data, training, model and prediction - Machine Learning and its three types: supervised, unsupervised and reinforcement learning - Deep Learning and neural networks (basic idea)	Training a simple image recognition model with Google Teachable Machine (<i>Demonstration</i>)
	1.3 Generative AI and Large Language Models	- Meaning of Generative AI - Generation of text, images, audio, video and code - Large Language Models and next-word prediction - Strengths and limitations of AI; AI hallucination	- Asking a chatbot the same question twice and comparing the answers - Finding a wrong AI answer and checking it against the textbook
MODULE 2 AI and Society	2.1 Applications of AI	- AI in education: personalised learning and doubt solving - AI in healthcare: reading X-rays and eye scans - AI in agriculture: crop disease detection and weather advisories - AI in banking, transport and public services - AI initiatives in India: IndiaAI Mission and Bhashini	Group discussion on one AI application from each sector
	2.2 Risks, Ethics and Careers	- Deepfakes and misinformation - Data privacy and bias in AI systems - Over-dependence on AI - Impact of AI on jobs, and new careers such as AI/ML engineer, data analyst and AI product designer - Skills needed for the future	“Thumbs up or thumbs down” debate on statements about AI - Career mapping activity

Module	Chapter	Topics Covered	Practicals
MODULE 3 Prompt Engineering with Gemini and Gemma	3.1 Basics of Prompting	- What a prompt is - Structure of a good prompt - The R-T-C-F method: Role, Task, Context and Format - Common mistakes in prompting	Rewriting a weak prompt (“Write an essay on pollution”) using R-T-C-F and comparing the two answers in Gemini
	3.2 Using Gemini for Learning	- Features of the Gemini app: chat, file and image upload, and Canvas - Follow-up prompts - Using AI as a tutor that gives hints instead of answers - Explanations in Hindi and other Indian languages	- Explaining a science concept at Class 8 level and generating three practice questions - Getting step-by-step hints for a mathematics problem
	3.3 Open Models and Gemma	- Cloud AI and on-device AI - What open models are - The Gemma family of models - Privacy and offline use of AI	Running Gemma offline on a mobile phone through the Google AI Edge Gallery app (<i>Demonstration</i>)
MODULE 4 AI for Learning with NotebookLM	4.1 Introduction to NotebookLM	- Source-grounded AI - Adding sources: PDFs, Google Docs, websites and YouTube videos - Citations and verification of answers	Creating a notebook from a textbook chapter and asking questions with citations
	4.2 Study Tools in NotebookLM	- Study guides and briefing documents - Flashcards and quizzes - Mind maps - Audio Overview and Video Overview - Using NotebookLM for revision	- Generating flashcards, a quiz and a mind map from the chapter “Life Processes” (Class 10 Science) - Creating an Audio Overview in Hindi
MODULE 5 AI for Images and Presentations	5.1 Image Generation with Nano Banana	- Text-to-image generation - Nano Banana and Nano Banana Pro in the Gemini app - Writing image prompts: subject, style, colour and layout - Editing images through text instructions - Acknowledging AI use, and the SynthID watermark	Designing a “Save Water” campaign poster and editing it with follow-up instructions
	5.2 AI for Presentations and Infographics	- Drafting presentations with Gemini Canvas - Exporting a presentation to Google Slides - Creating infographics and slide visuals - Slide Deck in NotebookLM	Preparing a seven-slide presentation on “Natural Disasters in India and How to Stay Safe”

Module	Chapter	Topics Covered	Practicals
MODULE 6 AI for Video Creation with Google Flow	6.1 Introduction to AI Video Generation	- How AI generates video - Google's Veo model - Features of Google Flow: Text to Video, Frames to Video, Ingredients to Video and Scene Builder	Creating an eight-second clip of a boat ride through the Marble Rocks at Bhedaghat (<i>Demonstration</i>)
	6.2 Video Prompts and Storytelling	- Elements of a video prompt: subject, action, setting, camera movement, lighting and sound - Planning a short story scene by scene - Use of AI video in school projects - Identifying deepfake videos	Writing a three-scene storyboard for an awareness video
MODULE 7 Building Apps with AI using Lovable	7.1 Introduction to AI App Builders	- Websites and web apps - How AI converts plain language into code - Working with Lovable: describe, generate, preview, refine and publish - Role of logic, testing and coding	Building a quiz website on "Our Solar System" for Class 9 students (<i>Demonstration</i>)
	7.2 Planning and Improving an App	- Breaking an idea into features - Writing clear instructions for the AI - Testing and fixing errors - Careers in software development	Planning a simple app, such as a portfolio, class timetable or study planner, and listing its features on paper
MODULE 8 Automation with Google Workspace Studio	8.1 Basics of Automation and AI Agents	- Meaning of automation - Trigger, Steps and Result - Difference between a chatbot and an AI agent - Automation in daily life	Designing an automation for a daily task in the Trigger, Steps and Result format
	8.2 Automation with Google Workspace Studio	- Overview of Google Workspace Studio - Creating agents by describing a task in plain language - Working across Gmail, Drive, Docs, Sheets, Calendar and Chat - Need for human supervision	- Recording science exhibition registrations from a Google Form in a Sheet and sending confirmation emails automatically (<i>Demonstration</i>) - Weekly email summary posted as a to-do list (<i>Demonstration</i>)

Module	Chapter	Topics Covered	Practicals
MODULE 9 Safe and Responsible Use of AI	9.1 Digital Safety and Privacy	- Protecting personal information - Safe use of AI apps - Age limits and guidance from parents and teachers - Cyber crime and its reporting: helpline 1930 and cybercrime.gov.in	Case discussion on what is safe and unsafe to share with an AI tool
	9.2 Academic Integrity and Ethics	- Fact-checking AI answers - Using AI to learn, not to copy - Acknowledging the use of AI in assignments - Respect for others: no deepfakes, photo morphing or bullying	- Fact-checking an AI answer with the textbook - Preparing a classroom code of conduct for using AI

4. COURSE OUTCOMES

After completing this programme, students will be able to:

- CO1 Explain the basic concepts of AI, Machine Learning and Generative AI, and identify AI in daily life.
- CO2 Discuss the benefits and risks of AI for society, education and careers.
- CO3 Write clear and effective prompts to get useful answers from AI tools.
- CO4 Use AI tools for studying and for creating images, presentations and videos.
- CO5 Understand how apps and automations are built with AI, and why logic and coding still matter.
- CO6 Use AI safely, honestly and responsibly.

5. TOOLS COVERED

S.No.	Tool	Use in the Programme
1	Gemini	Asking questions, explanations and presentations
2	Gemma	Open AI model that works offline on a phone or laptop
3	NotebookLM	Studying from textbook chapters and notes
4	Nano Banana	Creating and editing images, posters and slides
5	Google Flow	Creating short AI video clips
6	Lovable	Building websites and apps from plain language
7	Google Workspace Studio	Automating everyday tasks with AI agents

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Date: 05/10/2026

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